

Patent timing and industry ordering: does patenting lead trademark vocabulary?

Eric Silver

July 25, 2026

A patent is applied for before an invention is in public use; a trademark is filed when the firm is ready to sell. If ΔKL and patenting tracked one underlying event, patents should therefore *lead* trademark language. Table 1 enters patents at four timings in the paper’s within-firm specification. Table 3 orders the NICE classes by the lead coefficient.

Table 1: Within-firm coefficient of ΔKL on $\log(1 + \text{patents})$ at four patent timings. Firm and year fixed effects, standard errors clustered on owner, owners with at least three years. ΔKL is standardized once on the contemporaneous panel, so all coefficients are in the same units as the published -0.010 . Topic scoring, $T = 200$.

Patent timing	Coef. (σ)	SE	t	Owner-years	Owners
Patents at $t - 2$	-0.0149	0.0093	-1.61	40,211	6,244
Patents at $t - 1$ (patents lead)	-0.0092	0.0085	-1.08	47,224	7,423
Patents at t (contemporaneous)	-0.0095	0.0063	-1.51	89,987	15,121
Patents at $t + 1$ (reverse-timing placebo)	$+0.0026$	0.0079	$+0.33$	53,520	8,355

Table 2: The same lead specification ($t - 1$) inside the a-priori IPR groups, and the pooled interaction. The firm fixed effect absorbs the level of the firm-constant complement dummy, so the interaction is identified off the slope difference alone.

Stratum	Coef. (σ)	SE	t	Owner-years	Owners
Complement sectors	-0.0003	0.0107	-0.03	30,468	4,767
Neutral sectors	-0.0176	0.0209	-0.85	7,791	1,258
Substitute sectors	-0.0260	0.0180	-1.45	8,965	1,398
$\log(1 + \text{pat}) \times \text{complement}$	-0.0083	0.0171	-0.49	47,224	7,423

Table 3: NICE classes ordered by the lead ($t - 1$) coefficient.
C = a-priori complement, S = substitute, N = neutral. Classes with fewer than 50 owners are omitted.

Class	Group	Coef. (σ)	t	Owner-years	Owners
029 Meat/Foods	S	$+0.1966$	$+1.50$	305	51
008 Hand tools	N	$+0.0915$	$+0.86$	411	72
003 Cosmetics	N	$+0.0433$	$+1.49$	1,490	202
006 Metals	N	$+0.0329$	$+0.40$	740	135
019 Building	N	$+0.0324$	$+0.48$	592	107
017 Rubber/Plastics	C	$+0.0163$	$+0.18$	515	92
030 Coffee/Staples	S	$+0.0110$	$+0.20$	664	97

Class	Group	Coef. (σ)	t	Owner-years	Owners
005 Pharma	C	+0.0105	+0.36	4,090	624
035 Advertising/Retail	S	+0.0096	+0.25	778	138
020 Furniture	N	+0.0053	+0.09	1,108	182
012 Vehicles	C	+0.0047	+0.11	1,953	281
009 Software/Electronics	C	+0.0033	+0.21	12,467	1,953
036 Insurance/Finance	S	-0.0002	-0.01	743	102
010 Medical devices	C	-0.0019	-0.07	3,479	527
011 Lighting/HVAC	C	-0.0066	-0.16	1,949	313
014 Jewelry	S	-0.0112	-0.17	336	53
025 Clothing	S	-0.0113	-0.24	1,151	193
041 Education/Entertainment	S	-0.0129	-0.34	1,511	215
044 Medical services	N	-0.0330	-0.48	337	54
028 Toys/Games	S	-0.0354	-0.54	1,879	290
042 Scientific/Tech services	C	-0.0514	-1.23	1,102	202
001 Chemicals	C	-0.0700*	-1.84	2,065	323
038 Telecom	N	-0.0760	-0.83	358	65
007 Machines	C	-0.0897*	-1.79	2,459	388
031 Agriculture	S	-0.1146*	-1.83	453	67
016 Paper	N	-0.1380***	-2.77	1,018	168
021 Housewares	N	-0.1415**	-2.51	1,034	151

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Every owner in the panel is matched to a PatentsView assignee and so patents at least once; the substitute cells are patenting firms in low-patenting sectors rather than typical firms in those sectors, which makes the two groups more alike and biases the interaction toward zero.

Provenance. This note was produced with AI assistance. The author specified the research question, the timing logic, the estimating specification, and the a-priori sector grouping, and is responsible for its contents. Claude (Anthropic), model Claude Fable 5, wrote `scripts/patent_timing_by_industry.py`, ran the estimations, and drafted this note. Nothing here rests on that tooling: the script, the panel build, and the underlying corpus are public at <https://github.com/ericssilver/novelty>, so every coefficient in these tables can be reproduced independently.